

share it with the world!” Jeremiah Harmsen from Google Zurich excitedly explains. He adds that TFHub gives people ‘smaller pieces’ of machine learning models, making it more likely for developers to reuse code.

It is true that models require a specific type and format of input to produce a specific output. The minute you try to modify a model, you’re likely to run into issues. With TFHub, models and modules are treated as analogues to binaries and libraries. With this, it becomes well-defined in terms of importing modules and using functionality from within it. Modules are then composable, reusable, and retrainable.

In a talk at the TensorFlow Developer Summit, researchers at Google Zurich explained how you can customise a TFHub module for your own application, by using the example of an image classification model. Most image classifiers today have already been trained for hundreds of hours on millions of images. However, if given a custom task of making predictions related to a few types of rabbits, it is not possible to find a model that directly handles this for you. It becomes a complex problem to start to solve from scratch, but with TFHub and three lines of code, you find yourself retraining a model that can give you much better results than you would probably have obtained had you started from scratch. This approach could also incorporate the weights and architectures of models that have taken researchers over 60,000 hours of effort to develop!

TFHub aims to foster engagement within the community and promote more transparency in machine learning. At the same time, it provides a one-stop shop for all models of TensorFlow; just add a few lines of code and you have a model running inference on new input data or better still, re-training on input data to provide an even better, more fine-tuned performance. As expected, Google remains committed to pushing out more models and enabling a better Web experience on TFHub, as

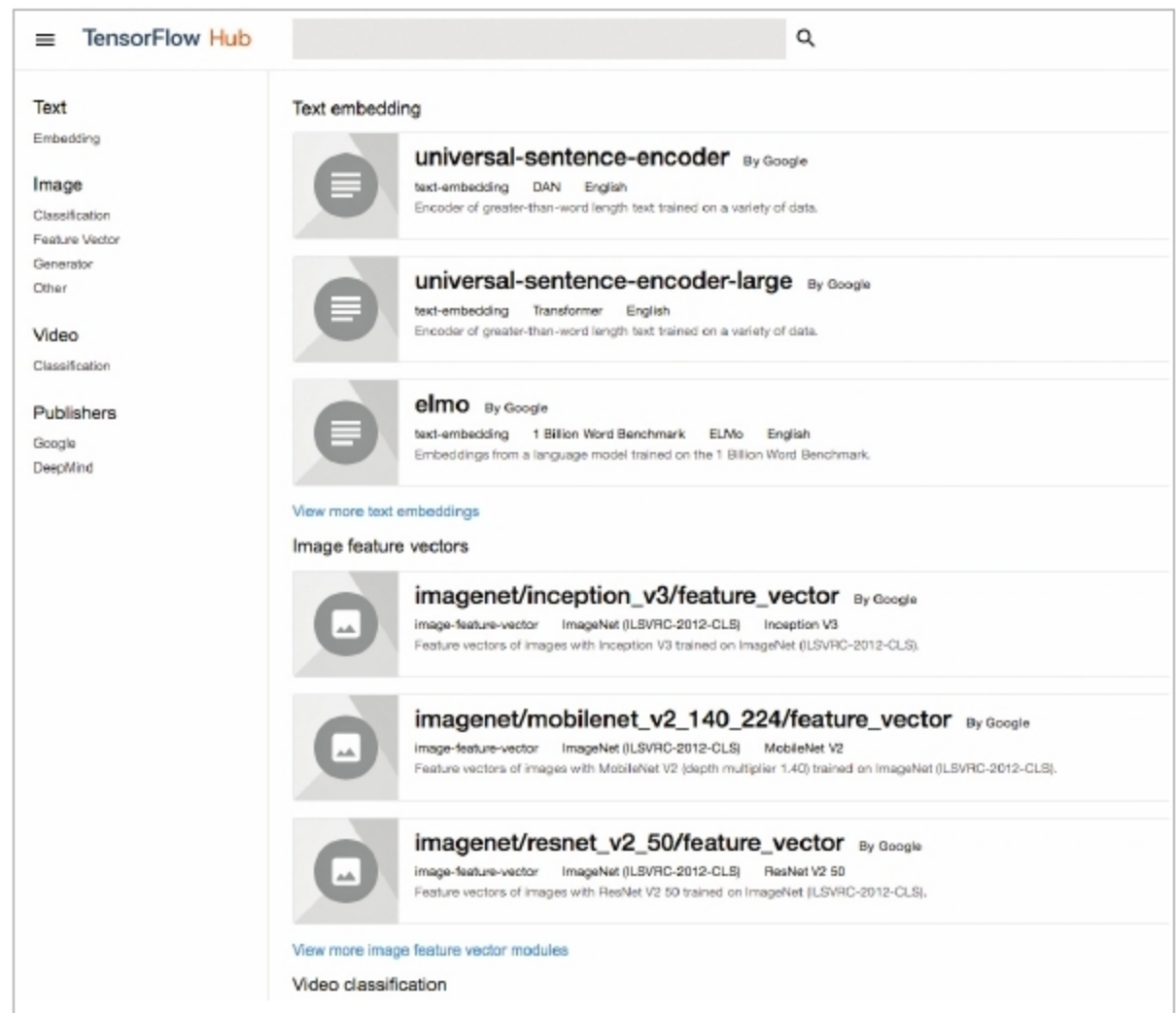


Figure 4: Take a look at the TFHub Web interface!

evidenced by the focus on providing resources akin to the ‘Universal Sentence Encoder’ module –“a successful example of speeding up the translation from fundamental machine learning science to application in the broader developer community.”

For many teams undertaking cutting-edge research, it is not feasible to publish their TFHub modules publicly. In such cases, TFHub allows developers to publish modules to, and consume from, private storage. Instead of referring to modules by their *tfhub.dev* URL, the User Guide explains how you can use a file system path to retrieve and use them in your code.

On the whole, with the introduction of TensorFlow 2.0 and the infrastructure offered by TensorFlow Hub, Google is doubling down on its commitment to provide developers with a complete

package, along with bells and whistles, to perform cutting-edge machine learning. Other projects in this area include TensorFlow data sets that capitalise on the *tf.data* API in order to allow for easy pipelining and data manipulation, *Tensorflow.js* which offers in-browser training of neural networks among other awesome functionality, and *Tensorboard* which is a visualisation tool for machine learning.

TFHub has been surrounded by a lot of hype ever since its introduction, but it remains to be seen if the product is used as widely as it was expected to. The Google Zurich team is a fantastic team of engineers and researchers, who are working in tandem on TFHub, in what Dr Harmsen states is its first major infrastructure project. There are big things coming out of this lab, so it is going to be very exciting to see what is next! **END** 🐧

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